

Oxygen That Needs a Mask

Perfluorocarbons carry oxygen by a completely different physical rule than hemoglobin does, and the difference explains why they were both wildly promising and awkward to use

A hemoglobin-based carrier is a chemical solution to a physical problem. Oxygen barely dissolves in water, so blood evolved a molecule that binds it. Everything about the BHOC story, from the affinity curve to the iron chemistry to the trouble with nitric oxide, follows from that fact.

There is another way to carry oxygen, and it involves no binding at all. Certain synthetic liquids, chains and rings of carbon in which nearly every hydrogen has been replaced by fluorine, dissolve oxygen the way a sponge soaks up water. They are called perfluorocarbons, and they have a long and peculiar history as oxygen carriers. This essay lays out how they work, where they fit, and what the comparison teaches about the hemoglobin products that dominate the conversation on this site.

The rule of the sponge

When oxygen dissolves in a liquid, the amount that dissolves is proportional to its partial pressure. This is Henry's law, and it is one of the simplest relationships in physical chemistry. Double the pressure and you double the dissolved oxygen. In ordinary plasma, the solubility is low: at the pressure of arterial blood, a deciliter of plasma holds only a fraction of a milliliter of dissolved oxygen.

A perfluorocarbon liquid dissolves far more, on the order of twenty times more than water by the figures usually quoted, and a pure liquid of this type can hold something like forty or fifty milliliters of oxygen per hundred milliliters when exposed to pure oxygen at atmospheric pressure. I offer those as rough figures from the literature, and they vary by compound.

But the rule is a straight line. Hemoglobin follows an S-shaped curve: it loads almost fully at the pressure of ordinary lung air and then holds its oxygen tightly until the pressure falls near that of the tissue. A perfluorocarbon does neither. It picks up oxygen in proportion to how much is around and gives it up in proportion to how much is missing. The consequence is the defining feature of the whole class: to load a perfluorocarbon with a useful amount of oxygen, you need a high oxygen pressure. The patient must breathe oxygen-enriched air or oxygen. That is what I mean by needing a mask.

This is not a small caveat. It is the opposite of hemoglobin's design. A hemoglobin carrier can be loaded by breathing room air. A perfluorocarbon carrier needs an oxygen supply and a delivery apparatus in addition to the infusion, which is a logistical and clinical complication in the very settings where a carrier is most wanted.

Turning a liquid into an injectable

Perfluorocarbons are hydrophobic and lipophobic, meaning they mix with neither water nor oil. You cannot inject them as they are. Making an intravenous product requires turning the

liquid into a fine emulsion, tiny droplets suspended in water with the help of an emulsifier. The droplets in these products are on the order of a fifth of a micrometer across, smaller than a red cell by a factor of about thirty.

That small size is what allows the droplets to travel through capillaries, and in principle to reach tissue that red cells cannot. It is the same rationale that motivates the hemoglobin carriers, achieved by a different route.

The emulsions carry their own problems. Emulsions can be unstable, and the early ones had to be stored frozen, and thawed before use. The droplets are removed from the blood by cells of the reticuloendothelial system, mainly in the liver and spleen, so the product has a half-life measured in hours to days and leaves behind a burden of fluorinated material that is slowly exhaled over weeks to months. Recipients of early products commonly experienced flu-like symptoms, a reaction thought to be related to uptake of the droplets by macrophages and the release of inflammatory signals.

A product that reached the market

The most notable perfluorocarbon product was called Fluosol, developed in Japan and approved in the United States around 1989 for use in a specific setting: protecting heart muscle during balloon angioplasty, when a vessel is briefly blocked. The idea was clever. During the few minutes that a balloon was inflated in a coronary artery, blood could not pass, but the small droplets could squeeze past the balloon through collateral channels, and if the patient was breathing high concentrations of oxygen, they carried a useful amount to the starved heart tissue.

The product was subsequently withdrawn from the market. The reasons were commercial and practical as well as clinical: it needed to be stored frozen, prepared with several components before use, and required oxygen breathing, while advances in angioplasty equipment, such as balloons that allow blood to flow past, reduced the need for it. The details of its commercial life deserve better sourcing than I have gathered here, so I will keep to the shape of the story. It was approved, it was used, and it faded.

A later generation of emulsions, made with a different compound and stabilized with a phospholipid, reached late-stage clinical trials in surgical patients. One of them, developed under the name Oxygent, was tested in surgery to reduce the need for blood transfusion, and the program encountered safety concerns in a trial and did not reach the market. I mention these to make a general point about the class. A physically elegant idea did not translate into a successful product. The reasons were mostly practical, and practical reasons are where the field lives.

What hemoglobin has that the sponge lacks

Set the two designs side by side and each shows what the other lacks.

Hemoglobin gives a lot of oxygen capacity at low oxygen pressure. A concentrated hemoglobin solution holds many times the oxygen of an equal volume of plasma at ordinary arterial pressure, without needing the patient to breathe enriched oxygen. That is the great advantage of chemistry over physics.

The perfluorocarbon droplets, by contrast, hold oxygen in a way that responds smoothly to local conditions. In the lung, when the patient breathes oxygen, the droplets load; in the tissue, where pressure is low, they release. Because the relationship is linear, a perfluorocarbon delivers a large fraction of its oxygen in the tissue range, while hemoglobin at high affinity may not. In tissue that is very low in oxygen, the sponge gives up cargo readily.

The perfluorocarbon has no iron and therefore no oxidation problem. It does not scavenge nitric oxide, so it does not cause the vasoconstriction associated with free hemoglobin. It is not a biological protein, so it does not raise concerns about foreign proteins or prions. It can be stored, if the emulsion is stable, at room temperature.

The hemoglobin, meanwhile, has the S-curve, the ability to load in ordinary air, a greater carrying capacity per unit volume in the relevant pressure range, and the cooperative behavior that has been refined over hundreds of millions of years.

I find the pairing instructive because it shows that the problems of hemoglobin carriers are not properties of oxygen carrying in general. They belong to a specific chemistry. If a patient community worries that all oxygen carriers cause vasoconstriction, the perfluorocarbon record is a counterexample. If it worries that all oxygen carriers require a breathing apparatus, the hemoglobin record is a counterexample. The properties that define a class are chemical.

A worked example of how little a sponge carries

The straight-line rule can be turned into arithmetic, and the arithmetic is sobering. The figures below are round numbers chosen for illustration, not values for any product.

Start with ordinary blood. Hemoglobin at 12.5 grams per deciliter, fully saturated, holds about 17 milliliters of oxygen per deciliter, using the usual figure of 1.34 milliliters per gram. Plasma, by dissolved oxygen alone, adds about 0.003 milliliters per deciliter for each millimeter of mercury of pressure, so at an arterial pressure of 100 it adds roughly 0.3. Tissues normally extract about 5 milliliters per deciliter on a single pass.

Now suppose we infuse a perfluorocarbon emulsion until it makes up a tenth of the blood volume, and that the emulsion is 30 percent perfluorocarbon by volume, so that 3 percent of the blood is the liquid itself. Take a liquid that dissolves about 45 milliliters of oxygen per 100 milliliters at one atmosphere of pure oxygen. While the patient breathes room air, with an arterial pressure near 100 millimeters of mercury, the perfluorocarbon adds about 0.18 milliliters per deciliter. That is smaller than the oxygen dissolved in plasma and negligible against the 17 in hemoglobin.

Now let the patient breathe pure oxygen, raising arterial pressure to about 600. The perfluorocarbon load climbs to roughly 1 milliliter per deciliter. That is a six-fold increase, and it is meaningful for a patient whose hemoglobin has fallen very low. But it is still a small fraction of what healthy blood carries. The sponge does its best work when the patient has almost no red cells left, or when the flow is so poor that a little extra oxygen makes the difference.

The arithmetic explains both the appeal and the disappointment. It shows why perfluorocarbons were tried as a bridge for patients who refused blood, and why the amounts

involved could never replace a large hemoglobin deficit. It also shows why the concentration of the emulsion and the volume infused mattered so much, and why doses large enough to matter tested the limits of what patients tolerate.

The frozen product and the thaw

There is a practical detail of the first commercial emulsion that illustrates how much clinical adoption depends on mundane things. The product was supplied frozen, and it had to be thawed, combined with additional solutions, and prepared shortly before use. Hospital pharmacists and perfusionists had to schedule the preparation around the procedure. A drug that requires a checklist and a countdown is a drug that is used only when someone has planned for it.

The comparison with a room-temperature bovine hemoglobin bag makes the point vividly. A shelf-stable product can be grabbed and hung. It does not depend on a pharmacy. Convenience is not a footnote in emergency medicine. It is often the deciding factor between a product used and a product admired.

The oxygen-mask paradox

Now the paradox that gives the essay its title. If a patient in shock is already receiving high-flow oxygen through a mask, as is standard in emergency care, then a perfluorocarbon may load in the lung and be effective. The need for supplemental oxygen is not an extra burden in a patient who would receive it anyway.

The complication arises in the field. A soldier or an accident victim in a remote place has no oxygen cylinder, or has only a limited supply. The very setting that argues for a room-temperature, shelf-stable carrier, the field, is the setting where the extra requirement is hardest to meet. And in a patient who breathes room air, the droplets add little.

There is a second complication in the opposite direction. High concentrations of oxygen are not benign. Prolonged breathing of very high oxygen levels can injure the lungs, and it changes the balance of reactive oxygen species in tissue. A treatment plan that depends on a high inspired oxygen fraction inherits those risks.

The paradox teaches something about how to evaluate any carrier. Ask what the carrier requires in order to work. Hemoglobin requires that the lung function and that the iron remain reduced. Perfluorocarbons require enriched oxygen. Encapsulated hemoglobin requires that the vesicle remain intact. Each dependency is a potential failure point in a particular setting.

A different niche: where oxygen pressure is already high

The oddest and most encouraging uses of perfluorocarbons have been in places where the mask problem does not arise, because the oxygen is delivered directly.

One is organ preservation. In some preservation methods, an organ is stored in a two-layer arrangement in which a perfluorocarbon layer at the bottom is bubbled with oxygen and the organ, usually a pancreas, floats at the interface, receiving oxygen by diffusion. The perfluorocarbon acts as an oxygen reservoir, with no need for a patient to breathe anything.

This method has been studied in pancreas preservation for islet cell isolation, and the results have been described as encouraging in some studies and modest in others. I raise it because it demonstrates a use in which the sponge is well suited: an organ at low temperature and low metabolic demand, in a bath oxygenated from outside. This is exactly the scenario described in a companion essay on cold oxygen, and it shows that the perfluorocarbon approach might have a natural home in preservation.

Another is eye surgery, where certain perfluorocarbon liquids are used as a temporary tool to flatten a detached retina. That use has nothing to do with oxygen carrying, and it is a reminder that these compounds are chemically inert and safe enough for direct contact with delicate tissue.

A third, more speculative, is liquid ventilation, in which the lungs are partially filled with an oxygenated perfluorocarbon liquid. It was tested in premature infants and adults with lung injury, and it has been a favorite of science fiction. The clinical trials did not show the benefit that supporters hoped for. It nonetheless shows how far the field was willing to go.

The sponge in cold storage

Return to the cold. The physics of dissolved oxygen changes with temperature, and the change favors the sponge. Gas solubility in liquids increases as temperature falls. A perfluorocarbon at four degrees dissolves more oxygen than at body temperature. Hemoglobin, in contrast, grips oxygen more tightly in the cold, which makes it a poorer donor. In the cold, then, the relative merits reverse: the sponge gets better and the chemical carrier gets worse.

That observation, which follows directly from Henry's law and the temperature dependence of hemoglobin affinity, deserves attention from anyone thinking about preservation products. It suggests that a preservation fluid for an organ in the cold might do better with a physical carrier than a chemical one, or with a mixture in which each does the job it is best at.

I would go further. The most interesting design for a warm perfusion might combine both: a bovine hemoglobin carrier for capacity at ordinary oxygen pressures, and a perfluorocarbon component to buffer oxygen in the microvessels. I am not aware of such a combination being tested in the clinic, and I mention it as a question rather than a proposal.

Why the story matters to the patient reader

There are three points to take away for a patient audience.

First, oxygen carrying is not synonymous with hemoglobin. When someone says a BHOC is the only way to carry oxygen without red cells, they are wrong, and the history of perfluorocarbons is the proof. That matters because it places the BHOC in a competitive landscape. An honest advocate says which of the alternatives it is meant to beat, and on what.

Second, the history is a lesson in the difference between physical elegance and clinical success. Perfluorocarbons had beautiful physics and an ugly logistics profile. Hemoglobin has an ugly set of side reactions and a beautiful mechanism. Neither won. A patient community that understands both stories is less likely to be dazzled by a single clever property.

Third, it shows how much depends on the setting of use. Every one of the perfluorocarbon successes was a niche with special conditions: a balloon in a coronary artery, a bath for a pancreas, an eye. The general-purpose blood substitute remains unsolved. This does not mean that niche uses are failures. On the contrary, they may be the only path by which a technology reaches patients at all. A BHOC that finds a defensible niche, whether organ preservation, patients who cannot receive blood, or remote care, stands a better chance than one that aims at replacing blood.

One more observation belongs here, because it is easy to miss. The perfluorocarbon record is a reminder that regulators and clinicians judge a product by the whole system it lives in: the frozen box, the thaw, the oxygen tank, the mask, the monitoring. A carrier that is chemically brilliant and operationally clumsy loses to one that is chemically ordinary and operationally easy. That is worth remembering when the BHOC story is told in terms of molecules alone. The molecule is one part of the product. The bag, the label, the storage temperature and the training of the person who hangs it are the rest.

Questions to ask about any carrier

Borrowing from the comparison, I would suggest a short list for any product.

What does the patient have to breathe, and what does the site need to supply?

What is the carrier's half-life, and what happens to it when it leaves the circulation?

What is the physical form: a solution, an emulsion, a particle, and how stable is it on the shelf?

What does the product do to the tissues that clear it, since every carrier is cleared by something?

And what is the honest niche, the place where the property that limits the product is not limiting?

Those are ordinary engineering questions, and the perfluorocarbon field, for all its disappointments, taught the community to ask them.

Further reading

Reviews of perfluorocarbon-based oxygen carriers, including a 2010 review of products and trials and a review that traces the field from physics to physiology, describe oxygen solubility, emulsion properties and clinical experience in detail. Figures for solubility, droplet size and half-life vary by compound and formulation and are quoted here as rough orders of magnitude. The history of Fluosol's approval and withdrawal and of the Oxygent trial deserves confirmation in primary sources before it is repeated.